to the neutral element consists of the dimensions of the representations.

Here is, for example, the character table of S_3 :

S_3	Id	(12)	(123)
#	1	3	2
$\mathbb{C}_+$	1	1	1
$\mathbb{C}_-$	1	-1	1
$\mathbb{C}^2$	2	0	-1

It is obtained by explicitly computing traces in the irreducible representations.

For another example consider A_4, the group of even permutations of four items. There are three 1-dimensional representations (as A_4 has a normal subgroup $\mathbb{Z}_2 \oplus \mathbb{Z}_2$ and $A_4/(\mathbb{Z}_2 \oplus \mathbb{Z}_2) = \mathbb{Z}_3$). Since there are four conjugacy classes in total, there is one more irreducible representation of dimension 3. Finally, the character table is

A_4	Id	(123)	(132)	(12)(34)
#	1	4	4	3
$\mathbb{C}$	1	1	1	1
$\mathbb{C}_\epsilon$	1	ϵ	ϵ^2	1
$\mathbb{C}_{\epsilon^2}$	1	ϵ^2	ϵ	1
$\mathbb{C}^3$	3	0	0	-1

where $\epsilon = \exp(\frac{2\pi i}{3})$.

The last row can be computed using the orthogonality of rows. Another way to compute the last row is to note that $\mathbb{C}^3$ is the representation of A_4 by rotations of the regular tetrahedron: in this case $(123), (132)$ are the rotations by 120^0 and 240^0 around a perpendicular to a face of the tetrahedron, while $(12)(34)$ is the rotation by 180^0 around an axis perpendicular to two opposite edges.

Example 4.8.1. The following three character tables are of Q_8, S_4, and A_5, respectively: